

Week	Tuesday Class	Thursday Class	Suggested Readings
Module 1: ML1 Bridge			
25-27 August	Course Overview. Skill Assessment (for Group Formation)	Wrangling with features: categorical variables, missingness, scaling, transforming, selection.	
1-3 September	Mastering model evaluation and optimization	Introduction to PyTorch	
Module 2: Dealing with High Dimensional Data			
8-10 September	Regularized Regression through PyTorch and Stochastic Gradient Descent	Principal Component Analysis as a preprocessing step. Introduction to Pipelines	
15-17 September	Lab: Partial Least Squares Regression (PLS)	Support Vector Machine I: The Linear Case	
Module 3: Exploiting Non-Linear Relationships in Data			
22-24 September	Support Vector Machine II: The Non-Linear case through the Kernel Trick	Lab: Predicting Seismic Building Response with Support Vector Regrssion	A. Chapter 3: Maximum margin classification with support vector machines
29 September-1 Octobe	Kernels are EVERYWHERE	Lab: Non-linear dimensionality reductions: t-SNE, and UMAP	A. Chapter 3: Solving nonlinear problems using a kernel SVM
Module 4: Neural Networks			
6-8 October	Reading Day (No Class)	Neural Networks I: Introduction & Multi-Layer Perceptrons (MLPs)	
13-15 October	Lab: Manual implementation of MLPs for Classification and Regression using toy data	Neural Networks II: Deeper into model building, training and opmitization.	A. Chapter 11: Modeling complex functions with artificial neural networks
20-22 October	Lab: Application of Neural Networks to Real Data	Neural Networks III: Improving generalization	A. Chapter 13: Making model building more flexible with nn.Module
27-29 October	Lab: Implementing an autoencoder for non-linear dimensionality reduction	Bagging: Bootstrap Aggregating and Random Forest	
Module 5: Ensemble Methods & Meta-Learning			
3-5 November	Election day (No class)	Boosting: Adaptive Boosting and Gradient Boosting	
10-12 November	Lab: Classifying Internet of Things Device Types with Bagging and Boosting	Clustering I: k-Means and Gaussian Mixture Models	
Module 6: Tackling Population Heterogeneity			
17-19 November	Clustering II: Hierarchical Clustering and DBSCAN	Clustering III: Validation	
26-28 November	Lab: clustering	Thanksgiving (No class)	
1-3 December	Final Project Time		